

# Healthcare Data Analytics

Data Dictionary | Portfolio Project

## Dataset Profile

The source CSV contains **55,500 records** and **15 columns**. The original field names and categories are retained from the source dataset.

## Column Definitions

| Column             | Data Type | Analytical Role      | Description                                                                                                            |
|--------------------|-----------|----------------------|------------------------------------------------------------------------------------------------------------------------|
| Name               | object    | Identifier / text    | Patient name recorded in the source dataset. Treat as an identifying/text field; use cautiously in portfolio analysis. |
| Age                | int64     | Numeric              | Patient age in years at admission.                                                                                     |
| Gender             | object    | Categorical          | Patient gender category recorded in the dataset.                                                                       |
| Blood Type         | object    | Categorical          | Patient blood group.                                                                                                   |
| Medical Condition  | object    | Categorical          | Primary medical condition associated with the admission.                                                               |
| Date of Admission  | object    | Date                 | Date on which the patient was admitted.                                                                                |
| Doctor             | object    | Categorical / text   | Doctor associated with the admission.                                                                                  |
| Hospital           | object    | Categorical / text   | Hospital associated with the admission.                                                                                |
| Insurance Provider | object    | Categorical          | Insurance provider associated with the admission.                                                                      |
| Billing Amount     | float64   | Numeric / currency   | Billing amount recorded for the admission.                                                                             |
| Room Number        | int64     | Numeric / identifier | Room number associated with the admission.                                                                             |
| Admission Type     | object    | Categorical          | Admission category: Elective, Urgent, or Emergency.                                                                    |
| Discharge Date     | object    | Date                 | Date on which the patient was discharged.                                                                              |
| Medication         | object    | Categorical          | Medication recorded for the admission.                                                                                 |
| Test Results       | object    | Categorical          | Recorded test-result category: Normal, Abnormal, or Inconclusive.                                                      |

## Derived Analytical Fields

**Length of Stay** can be derived as the number of days between *Discharge Date* and *Date of Admission*. This will be used for admission-duration analysis and dashboard KPIs.

## Data Quality Checks for This Project

The project should explicitly document duplicate-record checks, date validation, missing-value checks, and billing-value validation before analysis. These checks will be implemented in the SQL/Python workflow rather than altering the raw source file.

## Important Portfolio Practice

The raw CSV should remain unchanged in the **dataset** folder. Any cleaned or transformed version should be stored separately and clearly labelled. This makes the analytical workflow reproducible and easier for a recruiter to inspect.